

Test Plan — VWO Login Dashboard Authentication

Field	Value
Test Plan ID	TP-VWO-AUTH-001
Project Name	VWO Login Dashboard — Authentication & Login Experience
Application Under Test (AUT)	VWO login dashboard at https://app.vwo.com/
Test Plan Version	0.9 (Draft)
Document Status	Under review — pending Project Manager approval
Author	QA Automation Tester (4.8 years experience)
Reviewers	Project Manager (<i>pending</i>), Dev Lead, DevOps, Product Owner
Approvers	Project Manager, Product Owner
Standards Reference	IEEE 829-1998, ISO/IEC/IEEE 29119-2, ISO 25010, OWASP ASVS 4.0, WCAG 2.1 AA
Basis of Testing	Product Requirements Document (PRD) — VWO Login Dashboard, 7 pages
Planned Start / End Date	[TBD – to be baselined after PM review]

1. Document Control

1.1 Change Log

Version	Date	Author	Change Description
0.9	2026-08-12	QA Automation Tester	Draft created from PRD; submitted for PM review (not baselined)

1.2 Distribution List

Recipient	Role	Purpose
Project Manager	Reviewer / Approver	Plan approval, milestone sign-off
QA Team	Execution reference	Test scenario and execution ownership
Dev Lead / DevOps	Environment & defect coordination	Fixing, environment provisioning
Product Owner	Requirement clarification / UAT	Requirement confirmation, acceptance

2. Introduction

2.1 Purpose

Define the testing goals, scope, strategy, resources, schedule, and acceptance conditions for the **VWO Login Dashboard** at <https://app.vwo.com/>. This Test Plan is the controlling document for all test activities planned against the **PRD — VWO Login Dashboard (7 pages)** and is submitted for **Project Manager review**. It is **not baselined**; estimates, timelines, and targets are to be confirmed at review.

2.2 Scope of the Document

Covers the planned test phases (component → integration → system → regression → acceptance), test types, entry/exit criteria, test environment, automation scope, defect management, metrics, and reporting for the login-dashboard scope defined in Section 5. It is a **planning document and contains no executable test cases** — those are maintained separately in the test case repository (Task_03, [TestCases_Login_VW0.md](#)).

2.3 References

Ref	Document / Standard
[R1]	Product Requirements Document — VWO Login Dashboard (7 pages, provided)
[R2]	IEEE 829-1998 — Standard for Software Test Documentation
[R3]	ISO/IEC/IEEE 29119-2 — Software Testing Standards
[R4]	ISO/IEC 25010 — Systems and software Quality Requirements and Evaluation
[R5]	OWASP ASVS 4.0 / OWASP Authentication Cheat Sheet
[R6]	WCAG 2.1 AA — Web Content Accessibility Guidelines
[R7]	Test Case Repository — <code>TestCases_Login_VwO.md</code> (98 cases)

2.4 Terms, Definitions & Abbreviations

Term	Definition
AUT	Application Under Test
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
IdP	Identity Provider (SAML/OAuth)
SSO	Single Sign-On
MFA / 2FA	Multi-Factor Authentication / Two-Factor Authentication
CDN	Content Delivery Network
S1–S4 / P0–P3	Defect severity / priority classifications (Sections 12.2–12.3)
WCAG	Web Content Accessibility Guidelines

3. Application Under Test (AUT)

3.1 System Overview

The AUT is the **VWO login dashboard** — the critical entry point to VWO (Visual Website Optimizer), a digital experience optimization platform used by over 4,000 brands across 90 countries. The dashboard provides a clean, minimalist login interface with standard authentication fields, Remember Me functionality, a free-trial registration path, and product announcements (including Light/Dark mode). It is the gateway to VWO’s experimentation, personalization, and analytics suite.

3.2 Test Items

ID	Test Item	Version/Build	Supplier
TI-01	VWO login page (UI form, validation, Remember Me, forgot password entry, theme support)	[TBD]	VWO Frontend
TI-02	Authentication API / session endpoints (login, token issuance, session timeout)	[TBD]	VWO Backend
TI-03	SSO / social login integration (SAML, OAuth — Google, Microsoft)	[TBD]	IdP / VWO Backend
TI-04	Password reset flow (token generation, email delivery, recovery)	[TBD]	VWO Backend

3.3 Test Data

- **Synthetic test accounts only — no production user data.**
- Valid / invalid email formats (format, length, case, Unicode).
- Boundary-length passwords, special characters, whitespace.
- Locked, disabled, and unverified accounts (via test tenant).
- Test SSO / Google / Microsoft identity provider accounts.
- 2FA-enabled and 2FA-disabled accounts.
- Password-reset tokens (valid, expired, reused, malformed).

4. Test Objectives & Quality Targets

Quality Attribute	Objective	Measurable Target	Source
Functional Correctness	All PRD functional requirements behave per specification	≥ 98% scenario pass rate; zero open S1 defects	[R1]
Login Success Rate	Successful authentication attempts	≥ 95% (KPI)	[R1]
Performance — Page Load	Login page load time on standard connection	≤ 2 seconds (PRD KPI)	[R1]
Availability	Global uptime support	99.9% high availability (verify & document)	[R1]
Security	No successful brute force / unauthorized access; OWASP-aligned	Zero S1 security findings at exit	[R1],[R5]
Compliance	GDPR / CCPA data handling; enterprise audit support	100% compliance-audit requirements at exit	[R1]
Accessibility	WCAG conformance	WCAG 2.1 AA compliance	[R1],[R6]
User Satisfaction	Login experience quality	≥ 90% satisfaction (UAT survey)	[R1]
Support Volume	Login-related support tickets	Reduced by 20% (post-release trend, out of test scope)	[R1]
Defect Leakage	No high-severity defect to production	0 S1/S2 defects leaked	This plan

5. Test Scope

5.1 In-Scope

Functional:

- **Primary Authentication** — email + password login with secure validation.
- **Session Management** — secure session handling with **configurable timeout periods**.
- **Multi-Factor Authentication** — **optional 2FA** support.
- **Single Sign-On (SSO)** — enterprise SSO integration (SAML, OAuth).
- **Social Login** — optional integration with Google, Microsoft, and other identity providers.
- **User Input Validation** — real-time validation on blur, email format verification, password strength indicators, clear error messages.
- **Password Management** — Forgot Password flow, secure token generation, multiple recovery options, enforced password complexity.
- **User Experience** — responsive/mobile-optimized design, touch-friendly controls, auto-focus, clickable labels, loading states.
- **Accessibility** — screen reader support (ARIA), high-contrast mode, full keyboard navigation.
- **Branding & Visual Design** — VWO design system consistency, Light and Dark Mode.

Non-Functional:

- **Security** — encryption (E2E for auth data transmission), secure password storage (hashing), secure session tokens, HTTPS enforcement, rate limiting / brute-force protection.
- **Performance** — page load ≤ 2 s, asset optimization (compressed images, minified CSS/JS), CDN integration, concurrency sanity.
- **Compliance** — GDPR, CCPA, enterprise audit readiness.
- **Accessibility** — WCAG 2.1 AA.
- **Integration** — VWO core platform (dashboard transition), analytics (login success/failure tracking), customer support integration, marketing tools.

5.2 Out-of-Scope

- Post-authentication product functionality (A/B testing, CRO, analytics dashboards, personalization features) — tested by platform team.
- Native mobile applications — testing limited to mobile *browsers*.
- Full-scale load testing beyond concurrency sanity (scalability targets are verify-and-document).
- Biometric authentication, adaptive (risk-based) authentication, and PWA — **future enhancements per PRD**, not in this release.
- Third-party IdP internal behavior (tested at the integration boundary only).
- Billing / subscription management.

6. Test Strategy & Approach

6.1 Test Levels (V-Model / IEEE 829 Phases)

Phase	Scope	Approach	Primary Owner
Component / Unit	Form components, validation logic, password-strength indicator	Code-level tests; QA reviews coverage	Developers
Integration	Login UI ↔ auth APIs; SSO/OAuth redirects; session cookie issuance; analytics events	Contract tests (REST) + E2E	QA + Dev
System	Full login feature set against PRD requirements	End-to-end functional execution	QA
Regression	All in-scope scenarios after each change	Re-run automated suite per release	QA
Acceptance (UAT)	Business sign-off on login flows	Scripted scenarios from RTM; satisfaction survey	Product Owner

6.2 Test Types & Techniques

Test Type	Technique	Primary Tools
Functional	Black-box: equivalence partitioning, boundary value analysis, error guessing, decision tables	Playwright, Postman/Newman
UI / Visual	Layout, branding, Light/Dark mode, responsive breakpoints, visual diff	Playwright visual regression
Cross-Browser / Device	Parallel execution on real devices	Playwright matrix + BrowserStack
API	Auth contract, status codes, error payloads, token handling	Postman/Newman, REST Assured
Security	Credential validation, rate limiting/brute force, session cookie flags, SSO/OAuth token validation, injection (SQLi/XSS)	OWASP ZAP, Burp Suite (optional)
Performance	Page load, auth API response, concurrency sanity	K6 / JMeter, Lighthouse
Accessibility	WCAG 2.1 AA spot checks (labels, contrast, keyboard, screen reader)	axe-core, Lighthouse, screen reader (NVDA/VoiceOver)
Usability / Exploratory	Unscripted testing, UX heuristics, mobile touch targets	Manual

6.3 Test Design Techniques

- **Equivalence Partitioning** — valid/invalid input classes (email, password).
- **Boundary Value Analysis** — min/max password lengths, edge-case emails.
- **State Transition** — session states: active → idle → expired → re-authenticated; password-reset token states.
- **Decision Table** — auth method × credential validity × account state × 2FA state.
- **Error Guessing / Exploratory** — heuristic risk-based scenarios on the login boundary.

6.4 Execution Flow

1. **Preparation** — environment, test data, smoke check.
2. **Smoke Testing** — core login path on full browser matrix.
3. **Functional Execution** — P0 → P1 → P2 per RTM (Section 15).
4. **Cross-Browser / Mobile Execution** — full scenario set on device matrix.
5. **Non-Functional Execution** — security, performance, accessibility, compliance checks.
6. **Regression & Sign-Off** — full suite re-run; exit criteria verified; PM approval.

6.5 Test Data Management

- Synthetic accounts only; no production data.
- Versioned datasets; API-driven seeding where available.
- Cleanup and masking policies owned by Test Environment Owner.

7. Entry & Exit Criteria

7.1 Test Preparation Phase

Entry Criteria	Exit Criteria
Test Plan approved by PM	Test scenarios and RTM baselined
Environment provisioned & stable	Test data created and verified
SSO/OAuth/2FA test accounts provisioned	Smoke test passed on all target browsers

7.2 Functional Testing (System Test) Phase

Entry Criteria	Exit Criteria
Build deployed to test environment	100% of P0/P1 scenarios executed
Smoke test passed	≥ 98% pass rate
Test data available	No open S1 defects; S2 workarounds approved

7.3 Cross-Browser / Mobile Phase

Entry Criteria	Exit Criteria
Functional testing stable	All scenarios pass on 4 desktop + 2 mobile platforms
Device cloud confirmed	No unresolved S1/S2 platform-specific defects

7.4 Non-Functional Testing Phase

Entry Criteria	Exit Criteria
Feature freeze in place	Security scan clean; zero Critical findings open
Performance baseline available	Page load ≤ 2 s; P95 auth API within baseline
—	WCAG 2.1 AA blockers fixed or formally accepted

7.5 Regression & Release Exit Phase

Entry Criteria	Exit Criteria
Defects fixed and verified	Full regression suite ≥ 98% pass
Code freeze in place	Zero open S1/S2 defects
UAT environment ready	UAT signed off by Product Owner
—	Test Summary Report approved; PM release sign-off

8. Test Environment

8.1 Hardware & Software

Component	Specification
Desktop Browsers	Chrome, Firefox, Edge (latest 2 stable); Safari (latest 2)
Mobile Browsers	iOS Safari, Android Chrome (real devices)
Operating Systems	Windows 11, macOS latest, iOS 16+, Android 13+
Viewports	1920×1080, 1366×768, 768×1024, 375×667, 390×844

8.2 Configurations

- Production-like test environment with valid HTTPS/TLS.
- Test tenant with provisioned accounts (valid, invalid, locked, unverified, 2FA on/off).
- Test SSO IdP (SAML + OAuth), Google/Microsoft test clients.
- Documented auth API endpoints for contract testing.
- Simulated slow-network / low-bandwidth profile for load-time verification.

8.3 Tools

Tool	Purpose
Playwright	UI automation, cross-browser execution
BrowserStack / Sauce Labs	Real-device cloud
Postman / Newman	API contract testing
OWASP ZAP	Security scanning
K6 / JMeter	Performance baseline, concurrency sanity
Lighthouse / axe-core	Accessibility & page-load checks
Jira / Xray	Test management, defect tracking
Allure / ReportPortal	Test reporting
Git + CI (GitHub Actions/Jenkins)	Automation pipeline

8.4 Access & Support

Environment owner, credentials, and support SLA for the test environment [to be defined by DevOps] .

9. Test Schedule & Estimation

Estimates are **draft and not baselined** — to be confirmed at PM review.

Phase	Activity	Effort (person-days)	Milestone
P1	Planning, scenario design, RTM	3	Test Plan approved
P2	Automation framework & smoke scripts	5	Framework ready
P3	Functional execution + regression	6	Execution report
P4	Cross-browser / mobile	4	Device matrix report
P5	Non-functional (security/perf/accessibility)	3	NFR report
P6	Defect fixes, re-testing	4	Defect closure
P7	Summary report & sign-off	2	Release sign-off

Total: ~27 person-days (draft; parallelizable across 2–3 resources).

Key Dependencies: environment availability, SSO/OAuth/2FA account provisioning, session-timeout configuration freeze, and PM approval of the plan.

10. Roles & Responsibilities (RACI)

Role	Test Plan	Test Design	Execution	Automation	Defect Fix	Acceptance
QA Lead / Automation Tester (Author)	A/R	R	R	R	—	C
Project Manager	A	C	I	I	—	A
QA Analyst / Manual Tester	C	C	R	I	—	C

Role	Test Plan	Test Design	Execution	Automation	Defect Fix	Acceptance
Developer (Frontend/Backend)	C	C	C	C	R	C
DevOps / Environment Owner	I	—	C	C	I	—
Product Owner	C	C	I	—	—	R
Security Specialist	C	—	R	I	—	—

R = Responsible, A = Accountable, C = Consulted, I = Informed

11. Test Automation Plan

11.1 Automation Scope

- **In-Scope:** core email/password login, validation & error-message assertions, Remember Me persistence, Forgot Password entry, SSO/OAuth redirect flows (with test IdPs), session-timeout behavior, cross-browser smoke/regression, visual regression (Light/Dark mode).
- **Manual:** penetration testing, exploratory & usability, screen-reader accessibility verification, complex 2FA flows.

11.2 Framework & Tooling

- Framework: Playwright (TypeScript) + Page Object Model.
- Data: environment-seeded synthetic accounts; API-driven setup.
- Reporting: Allure/HTML with failure artifacts (video, screenshot, trace).

11.3 CI/CD Integration

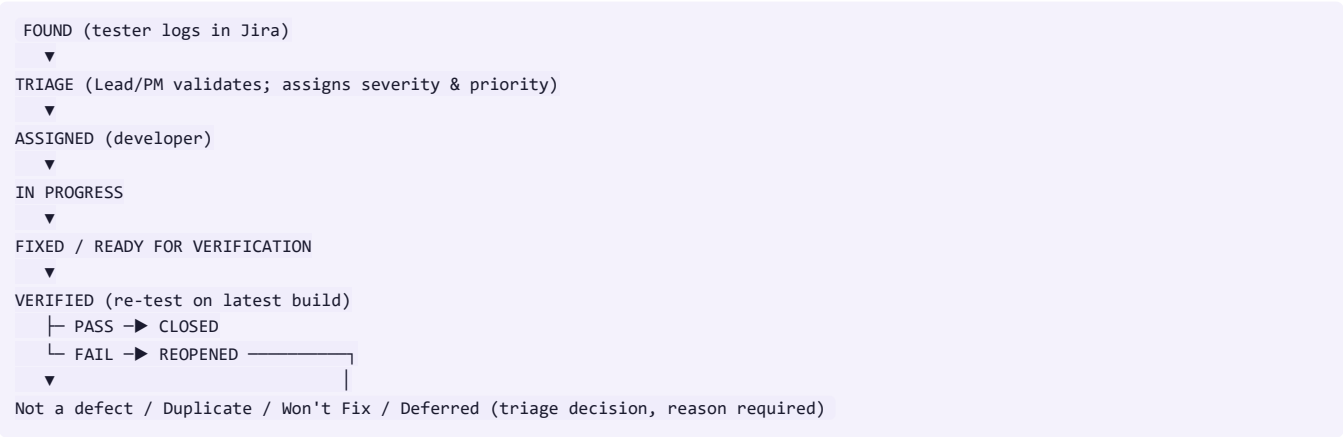
- Trigger: merge to test branch + scheduled nightly regression.
- Flaky-network retry policy (max 2 retries with logging).
- Anti-bot pacing between login attempts to avoid lockout/rate limiting.

11.4 Automation Governance

- Estimated coverage: 60–70% of the functional suite in the first cycle.
- Page objects shared; no test-logic duplication; code review required.

12. Defect Management

12.1 Defect Lifecycle



Rules: every defect includes title, environment, browser/OS, steps to reproduce, expected vs actual, severity, priority, attachments (logs, screenshots, HAR). Deferred defects require PM + PO sign-off.

12.2 Severity Classification

Severity	Label	Definition	Example
S1	Critical/Blocker	System unusable; security vulnerability; data loss; core auth failure	Login fails for all valid users; credential exposure; session hijack
S2	High	Major function broken, no workaround	Remember Me broken; SSO fails; password reset unusable
S3	Medium	Deviates from spec, workaround available	Wrong error message text
S4	Low	Cosmetic / UI polish	Alignment, spacing, minor theme contrast issue

12.3 Priority Classification

Priority	Label	Definition
P0	Urgent	Fix immediately; blocks release
P1	High	Fix in current sprint/release
P2	Medium	Fix in next release
P3	Low	Fix when convenient

12.4 Reporting Cadence

- Daily triage during execution; weekly summary to PM (open count by severity, aging, reopened rate); release-readiness review of open S1/S2 before exit.

13. Test Metrics & Reporting

Metric	Definition	Target
Execution Progress	% scenarios executed vs planned	100% by phase exit
Pass Rate	% passed vs executed	≥ 98%
Login Success Rate	Successful auth attempts (PRD KPI)	≥ 95%
Page Load Time	Login page load on standard connection (PRD KPI)	≤ 2 s
Defect Density	defects per scenario/requirement	[TBD]
Defect Age	open days per severity	[TBD]
Reopened Rate	% reopened after verification	< 10%
Automation Coverage	% automated vs total	60–70%
Defect Leakage	S1/S2 reaching production	0

14. Risk, Assumptions, Constraints & Dependencies

14.1 Risk Register

ID	Risk	Impact	Likelihood	Mitigation	Owner
R1	2FA/MFA flow not visible in current UI; configuration may vary by plan	Incomplete MFA coverage	Medium	Confirm 2FA configuration access; schedule as follow-up	QA Lead
R2	SSO/OAuth (Google, Microsoft) require external IdP test accounts	Environment delays	Medium	Pre-provision accounts; mock IdP for offline automation	DevOps
R3	Rate limiting / anti-bot may throttle automated login attempts	Automation flakiness	High	Test-tenant allowances; pacing between attempts	QA Automation
R4	Session timeout / Remember Me duration not documented	Ambiguous expected behavior	Medium	Freeze session config before execution	Product Owner

ID	Risk	Impact	Likelihood	Mitigation	Owner
R5	Light/Dark mode new feature may change layout/contrast	Visual regression risk	Medium	Include theme toggle in visual regression matrix	QA
R6	Availability/load targets (99.9%, concurrent users) not measurable in test	Scope gap vs PRD	Medium	Baseline sanity load; document as verify-and-document	Performance Tester

14.2 Assumptions

- Facts and stated assumptions below are tagged per the anti-hallucination protocol (Section 19). Items tagged **Inference** or **Insufficient information** must be confirmed before execution.

#	Assumption	Tag
A1	Test tenant with all account states (valid, invalid, locked, unverified, 2FA on/off) is provided by project team	Assumed — to be confirmed
A2	SSO, Google/Microsoft test accounts provisioned by DevOps	Assumed — to be confirmed
A3	"Configurable timeout periods" — exact session-timeout values are not published in the PRD	Insufficient information — verify & document
A4	"99.9% uptime" and "thousands of concurrent logins" are operational targets, not testable in QA env	Inference — verify & document
A5	2FA is optional per account; flow surfaces post-credential-verification (no 2FA UI on the initial screen per prior observation)	Inference (low confidence)
A6	Login success/failure analytics tracking is observable via a test analytics endpoint	Inference (low confidence)
A7	GDPR/CCPA compliance is validated via data-handling review, not full legal audit	Inference — verify & document

14.3 Constraints

- Test-only environments only; **no production user data**.
- Fixed release date [TBD].
- Limited device-cloud budget — prioritize matrix.

14.4 Dependencies

- PM approval of this plan.
- Environment readiness; external IdP provisioning.
- Session-timeout configuration freeze.
- Auth API contract availability.

15. Requirements Traceability Matrix (RTM)

Maps PRD requirements to scenario groups. **Individual test cases are maintained in the test case repository** (TestCases_Login_VwO.md , 98 cases); the matrix references scenario groups, not steps.

Req ID	PRD Requirement	Scenario Group	Priority	Phase	Verification Source
RQ-01	Primary authentication (email + password) with secure validation	TC-AUTH-01..05	P0	Functional + Regression	[PRD] Verified
RQ-02	Session management with configurable timeout	TC-SESS-01..04	P0	Functional + NFR	[PRD] Verified
RQ-03	Optional 2FA / MFA	TC-MFA-01..03	P1	Functional (follow-up)	[PRD] Verified; flow Inference
RQ-04	Enterprise SSO (SAML, OAuth)	TC-SSO-01..03	P0	Functional + Integration	[PRD] Verified
RQ-05	Social login (Google, Microsoft, others)	TC-GGL-01..02, TC-MSDN-01	P1	Functional + Integration	[PRD] Verified
RQ-06	Real-time validation on blur; email format verification; password strength indicator	TC-VAL-01..06	P1	Functional	[PRD] Verified

Req ID	PRD Requirement	Scenario Group	Priority	Phase	Verification Source
RQ-07	Clear, actionable error handling	TC-ERR-01..03	P1	Functional	[PRD] Verified
RQ-08	Forgot Password flow; secure token; multiple recovery options	TC-FP-01..04	P0	Functional	[PRD] Verified
RQ-09	Enforced password complexity	TC-PWD-01..03	P1	Functional + Security	[PRD] Verified
RQ-10	Remember Me persistent login	TC-RM-01..02	P0	Functional	[PRD] Verified
RQ-11	Responsive / mobile-optimized; touch-friendly; auto-focus; clickable labels; loading states	TC-UX-01..05	P1	Cross-Browser + Mobile	[PRD] Verified
RQ-12	Accessibility: screen reader (ARIA), high contrast, keyboard navigation	TC-ACC-01..04	P1	Accessibility	[PRD],[WCAG] Verified
RQ-13	Light and Dark Mode	TC-THEME-01..02	P2	UI/Visual	[PRD] Verified
RQ-14	Branding & visual consistency	TC-BRAND-01	P3	UI/Visual	[PRD] Verified
RQ-15	Security: E2E encryption, hashed storage, secure tokens, HTTPS enforcement, rate limiting	TC-SEC-01..06	P0	Security	[PRD] Verified
RQ-16	Performance: page load ≤ 2 s, asset optimization, CDN, concurrency sanity	TC-PERF-01..03	P1	Performance	[PRD] Verified
RQ-17	Compliance: GDPR, CCPA, enterprise audit readiness	TC-COMP-01..02	P1	Compliance review	[PRD] Verified
RQ-18	Integration: dashboard transition, analytics, support, marketing tools	TC-INT-01..03	P1	Integration	[PRD] Verified
RQ-19	Compatibility: Chrome/Firefox/Safari/Edge desktop; iOS/Android mobile	TC-COMPAT-01..03	P0	Cross-Browser + Mobile	[PRD] Verified

No orphan scenarios: every scenario group maps to a requirement, every requirement to a scenario group.

16. Test Deliverables

#	Deliverable	Owner	Due
1	Test Plan (this document)	QA Lead	PM approval
2	Test Scenarios & Case Repository (Task_03)	QA Team	Pre-execution
3	Test Data Sheet	Environment Owner	Pre-execution
4	Automated Test Suite	QA Automation	Framework ready
5	Test Execution Report (per phase)	QA Team	Per phase exit
6	Defect Report & Metrics	Test Lead	Weekly
7	Security Scan Report	Security Specialist	NFR phase
8	Performance Report	Performance Tester	NFR phase
9	Accessibility Report	QA Team	NFR phase
10	Cross-Browser/Device Matrix Report	QA Team	Cross-browser phase
11	Test Summary Report (with release recommendation)	Test Lead	Release exit
12	Sign-off Checklist	PM / PO	Release exit

17. Templates & Standards

- **Documentation standard:** IEEE 829-1998 aligned with ISO/IEC/IEEE 29119 terminology.
- **Test case template:** Requirement ID, scenario ID, priority, precondition, test data, steps, expected result, status, defect link (Jira-format per Task_03 repository).
- **Defect template:** Summary, environment, severity (S1–S4), priority (P0–P3), reproduction steps, expected vs actual, attachments.
- **Naming conventions:** Scenario IDs `TC-<AREA>-<NN>` ; defect IDs from tracker sequence.
- **Version control:** All test artifacts under Git; plan changes via version bump with change log.
- **Review cycle:** Test Plan → PM; Test Summary Report → PM & PO before release sign-off.

18. Approvals

Not yet approved — pending PM review. This section is intentionally unsigned.

Role	Name	Signature / Date
QA Lead (Author)	QA Automation Tester	[]
Project Manager	[]	[]
Product Owner	[]	[]
DevOps / Environment Owner	[]	[]

19. Appendix A — Verification & Anti-Hallucination Protocol

Every assertion in this plan and its downstream test cases is labelled with one of:

Label	Meaning
Verified	Directly traceable to a provided source (PRD, screenshot, API doc, run result) with a source tag, e.g., [PRD] , [DOM] , [RUN] , [SS]
Inference (low confidence)	Logical deduction that must be confirmed before execution
Insufficient information to determine	Cannot be asserted from available input — record as an open question, do not guess

Process (mandatory, deterministic):

1. Extract verifiable facts from the provided input only (this plan is built on the PRD).
2. List unknown or missing information.
3. Generate output strictly from Step 1 facts; label every inference.
4. Self-check for hallucinations or contradictions.
5. Configuration-dependent values (lockout thresholds, rate limits, session timeouts, HTTP status codes, cookie flags, uptime/concurrency targets) are recorded as *verify-and-document* checks, never asserted as fact.
6. If any step cannot be completed, stop and report why.

Self-Validation Check:

- ☐ All requirements asserted in this plan are traceable to the PRD.
- ☐ No error codes, endpoints, or features invented.
- ☐ Every inference labelled `Inference (low confidence)` .
- ☐ No contradictions between sections; discrepancies escalated as assumptions (Section 14.2).
- ☐ Missing information flagged as dependencies / open questions.

20. Appendix B — Open Items for PM Review (Not Final)

The following are **draft/open items** that the Project Manager should confirm before this plan is baselined:

#	Open Item	Requested Decision
O1	Test plan version/ID format and change-control process	Confirm <code>TP-VW0-AUTH-001</code> ; adopt version bump process
O2	Release date and sprint alignment	Provide target release date for scheduling
O3	Test environment availability and tenant provisioning	Confirm env owner and SLA
O4	SSO/Google/Microsoft test account provisioning	Confirm availability and timeline
O5	Session-timeout and Remember-Me duration values	Freeze configuration for test baseline
O6	2FA flow availability in test tenant	Confirm 2FA configuration access or defer
O7	Analytics/login-tracking observability in test	Confirm analytics endpoint visibility
O8	Availability (99.9%) and concurrency targets — testable scope	Confirm whether load testing is required
O9	Automation coverage target (60–70%)	Confirm investment and priority
O10	Accessibility verification depth (automated vs manual screen-reader)	Confirm scope and tools

End of Test Plan (Draft v0.9) — submitted for Project Manager review. Not final; not baselined.